# Xuyan Huang

alvin4huang@gmail.com | xuyanhuang.com | github.com/huangxuyan1

## Education

**Brown University, Providence, RI** Sep 2026 – Expected May 2028
M.S. in Computer Science; coursework: Deep Learning, Self-Supervised Learning & World Models

**The Ohio State University, Columbus, OH** Aug 2023 – May 2026
B.S. in Computer Science and Engineering, AI Specialization; Minor in General Psychology
**GPA: 3.930/4.000; Summa Cum Laude**
**Coursework:** Machine Learning, Computer Vision, NLP, Neural Networks
**Awards:** HackAI 2024, AI Club Track Champion; MakeOHI/O 2024, Human-Centered Computing Winner; Buckeye Programming Competition, 3rd Place (2024); ICPC North America Qualifier, OSU 9th Place (2023)

## Work Experience

**Summer Intern, Union Mobile Financial Technology, China** May 2025 – Jul 2025

- Trained and evaluated five voice and music model architectures for an AI-powered visual novel game engine.
- Fine-tuned ACE-Step; found phase issues causing stereo bass in base and fine-tuned outputs.
- Designed evaluation pipelines for generated speech and music, comparing models for the team's game-engine use cases.
- Authored LLM prompt templates to extract character and background descriptions from novel text; templates were deployed by the end of the internship.

## Research Experience

**Undergraduate Researcher, OSU Machine Learning as the Basis Lab** Sep 2025 – May 2026

- Curated, labeled, and quality-checked video and segmentation data for BALLER120, a 4,583-clip benchmark spanning 120 NBA players for studying motion cues in video re-identification.
- Investigated synthetic music detection using audio foundation models in an independent research project.

**Undergraduate Researcher, OSU Natural Language Processing Group** Aug 2024 – Oct 2024

- Identified and annotated risky behaviors by LLM-based web agents for an early-stage guardrail project later developed into the WebGuard benchmark.

**Undergraduate Researcher, OSU Computer Security Laboratory** May 2024 – Apr 2025

- Defined WeChat API search criteria for the lab's existing fingerprinting scanner; wrote hash-based script grouping and JSON summary code, and found a case later analyzed in the group's SaTS 2025 paper.

## Publication

Yingtie Lei, Fangxun Liu, Baicheng Wu, Colin Lee, Ziheng Zhang, Junke Yang, Zhiyuan Tao, **Xuyan Huang**, et al. "Probing Identity-Specific Motion Signatures: A Controlled Diagnostic Study." arXiv:2607.03633, 2026.

## Selected Project

**Modular – AI Coding-Agent Harness | C++20** Jun 2026 – Present

- Engineered a coding-agent harness with project context, ADRs, specs, dependency-aware tickets, validation gates, and plan-to-spec-to-implementation-to-review workflows.
- Built a typed audio/MIDI graph runtime with nested supernodes, graph persistence, and preallocated buffers for block processing.

## Additional Technical Experience

**Music Technology X Program, Tencent SSV** Jul 2023 – Aug 2023

- Trained VISinger2 and SoftVC-VITS on Tencent Cloud; both models achieved the highest mean opinion score among program participants.
- Developed an inference tool that enabled VISinger2 to synthesize singing from DiffSinger (`.ds`) files.
- Assisted with recording and preparing a custom singing dataset for model training.

**Buckeye AutoDrive, The Ohio State University** Mar 2024 – Jun 2024

- Integrated V2X into a vehicle perception pipeline and predicted egress lanes from V2X and sensor waypoints.

## Technical Skills

**Languages:** Python, C++, Java, Ruby **ML:** PyTorch, TensorFlow, Hugging Face, CUDA, Slurm
**Audio/Data:** librosa, FFmpeg, SQLite, JUCE, Ableton **Development:** Git, Linux, Ruby on Rails, Bootstrap